MATH 1075 - Section 10.3, 10.4 Worksheet

11/5/2026

1 Simplifying Radicals

Simplify the radicals. (Hint: $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$)

1. $\sqrt{x^{15}}$

2. $\sqrt[3]{x^{17}}$

3. $\sqrt{x^5 y^4}$

4. $\sqrt{63}$

5. $\sqrt[3]{54}$

6. $\sqrt{25ab^3}$

7. $\sqrt[3]{-16x^6yz^3}$

8. $\sqrt{\frac{y^5}{y}}$

2 Adding and Subtracting Radicals

Add or subtract the radical expressions, if possible.

1. $3\sqrt{5} + 6\sqrt{5}$

2. $3\sqrt[3]{xy} - 2\sqrt[3]{xy} + \sqrt[3]{xy}$

3. $2\sqrt{11} + 3\sqrt{13} + 5\sqrt{11}$

4. $\sqrt[4]{xy} + \sqrt[3]{xy}$

5. $5\sqrt{32} + 2\sqrt{50}$

6. $17\sqrt[3]{81} - 2\sqrt[3]{24}$

7. $7\sqrt[3]{x^4} - x\sqrt[3]{x}$

8. $\sqrt{50x^2}$